BLA-05 · COMPUTE, CONTAINERS AND DATABASES

Where Your App Runs and Where It Remembers

Amazon EC2 · ECS, EKS and Fargate · Amazon Lex and Connect · Databases you should know · Caching and graphs for GenAI

PART 02

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Amazon EC2

Virtual servers you choose, size and pay for.

Instance family	Good for
General purpose (M, T)	Web apps, small services
GPU (G, P)	Hosting and training models
Inferentia / Trainium	AWS chips for cheaper inference and training

Pricing	Best when
On-Demand	Short or unpredictable use
Savings Plans / Reserved	Steady use for 1–3 years
Spot	Work that can be interrupted — up to ~90% cheaper

Containers: ECS, EKS and Fargate

A container packages your app with everything it needs, so it runs the same everywhere.

Service	What it is	Choose it when
Amazon ECS	AWS's own container service	You want the simplest option on AWS
Amazon EKS	Managed Kubernetes	Your team already uses Kubernetes
AWS Fargate	Runs containers without servers	You don't want to manage EC2

💡 **Fargate isn't a third choice —** it's where ECS or EKS containers can run.

Two roles every ECS task has

Role	Used by	Lets it…
Task execution role	ECS itself	Pull the image from ECR, send logs to CloudWatch
Task role	Your code	Call AWS services — like Amazon Bedrock

🏢 **A container calls Bedrock and gets AccessDenied —** because the Bedrock permission was added to the execution role, not the task role.

Amazon Lex and Amazon Connect

Service	What it does
Amazon Lex	Builds chatbots and voice bots — understands what the customer wants
Amazon Connect	A cloud contact centre — phone lines, queues and agents

Lex word	Meaning	Example
Intent	What the customer wants	CheckOrderStatus
Slot	A detail it needs	orderNumber

🏢 A customer phones in → Lex asks for the order number and answers → still stuck? Connect passes them to a person.

Databases you should know

The trick is knowing which kind of data each one is for.

Service	Stores	Think of it as	GenAI use
DocumentDB	JSON documents	MongoDB-compatible	Flexible app data, vector search
ElastiCache	Data in memory	Very fast short-term memory	Caching answers and sessions
MemoryDB	In memory, kept durably	Fast and safe	Low-latency vector search
Neptune	Relationships (graphs)	A map of who connects to what	Graph-based retrieval

Caching for GenAI

Model calls are slow and cost money. A cache remembers answers you've already paid for.

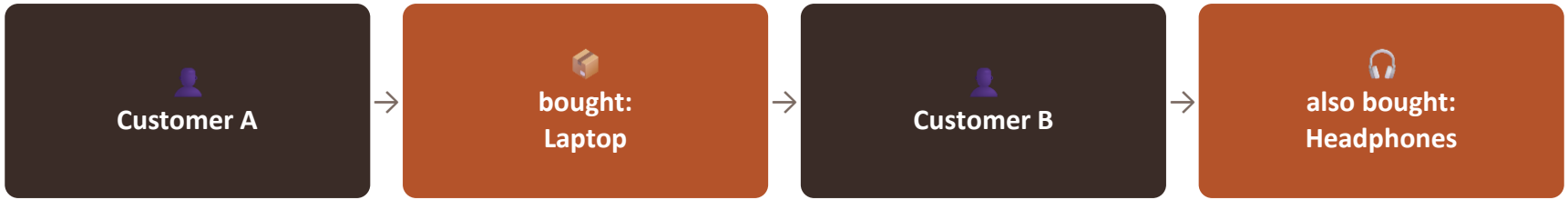
What to cache	Where
Chat session history	ElastiCache / MemoryDB
Answers to common questions	ElastiCache — a semantic cache
Embeddings you've already computed	ElastiCache / MemoryDB

🔑 **Valkey —** an open-source, Redis-compatible engine you can choose for both.

⚠️ **Give cached items an expiry time,** and never share one user's answer with another.

Graphs: Amazon Neptune

Some questions are about connections, not documents.



🏢 **"What do people who bought this laptop also buy?"** A graph answers it by following the links.

Neptune Analytics adds vector search to the graph — so RAG can use the meaning of text and the connections between things.

Key takeaways

1. Match EC2 pricing to the work — Spot for interruptible jobs
2. ECS for simplicity, EKS for Kubernetes, Fargate for no servers
3. Task role for your code, execution role for ECS
4. Match the database to the data: documents, memory or relationships
5. Cache what you've already paid for — with an expiry time

Next · Part 3: building it repeatably, and AI services you can just use